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Experiment 1: Case Study

Problem Statement: Choose a real-world case study on an E-Commerce System, create user stories, select an appropriate software development process model, identify a suitable programming language, and propose an implementation and evaluation strategy.

Real-World case study: A Multi-Vendor E-Commerce System

1. Functional Requirements

Customer

- Register and login
- Search and filter products
- View product details
- Add products to cart and wishlist
- Place orders
- Make online payments
- Track order status
- View purchase history
- Submit product reviews and ratings

Seller / Vendor

- Register and login as a seller
- Add/list new products
- Update product details and prices
- Remove or deactivate products
- Update product stock and availability
- View orders containing their products
- Update order/fulfillment status
- View sales and revenue reports

Admin

- Manage products
- Manage categories
- Update inventory
- View customer details
- Process and manage orders
- Manage discounts and offers
- Generate sales reports

2. User Stories

Customer User Stories

USER ID	User Story	Priority
CU-01	As a customer, I want to register and	High

	login so that I can securely access my account.	
CU-02	As a customer, I want to search and filter products so that I can quickly find the items I need.	High
CU-03	As a customer, I want to view product details so that I can make an informed purchase decision.	High
CU-04	As a customer, I want to add products to my cart so that I can purchase multiple items together.	High
CU-05	As a customer, I want to make online payments so that I can complete my order conveniently.	High
CU-06	As a customer, I want to track my order so that I know its current delivery status.	High
CU-07	As a customer, I want to view my purchase history so that I can review previous orders.	Medium
CU-08	As a customer, I want to submit product reviews and ratings so that I can share my experience with other customers.	Medium
USER ID	User Story	Priority
SU-01	As a seller, I want to register and login so that I can securely manage my products.	High
SU-02	As a seller, I want to add products so that customers can purchase them from the platform.	High
SU-03	As a seller, I want to update product details and prices so that the catalog remains accurate.	High
SU-04	As a seller, I want to update product stock so that customers can see product availability.	High
SU-05	As a seller, I want to view orders for my products so that I can process them efficiently.	High
SU-06	As a seller, I want to update order status so that customers can track their orders.	High
SU-07	As a seller, I want to view sales reports so that I can monitor my business performance.	Medium

Seller / Vendor User Stories

Administrator User Stories

USER ID	User Story	Priority
AUS-01	As an administrator, I want to manage products so that the online catalog remains accurate.	High
AUS-02	As an administrator, I want to manage categories so that products are organized properly.	High
AUS-03	As an administrator, I want to update inventory so that product availability is maintained accurately.	High

AUS-04	As an administrator, I want to process customer orders so that purchases are fulfilled on time.	High
AUS-05	As an administrator, I want to manage users and permissions so that system security is maintained.	High
AUS-06	As an administrator, I want to generate sales reports so that business performance can be monitored.	High
AUS-07	As an administrator, I want to manage discounts and offers so that promotional campaigns can be controlled.	Medium

3. Selected Software Development Process Model

Agile Scrum

Reason for Selection

- E-commerce requirements change frequently based on customer and business needs.
- Continuous feedback can be obtained from customers, administrators, and business stakeholders.
- Modules can be delivered incrementally through short sprints.
- Faster development cycles with regular sprint reviews.
- Easy bug fixing and adaptability to changing market requirements.
- Well-suited for web-based and mobile-friendly e-commerce applications.

Sprint Plan

Sprint	Features
Sprint 1	Customer, Seller & Admin Registration/Login
Sprint 2	Product Catalog, Categories & Seller Product Listing
Sprint 3	Product Search, Shopping Cart & Wishlist
Sprint 4	Order Management, Seller Order View & Checkout
Sprint 5	Payment Gateway & Seller Sales/Revenue Tracking
Sprint 6	Order Tracking, Seller Fulfillment & Customer Reviews
Sprint 7	Admin Dashboard, Inventory, Seller Management & Sales Reports
Sprint 8	Testing (Unit, Integration, UAT) & Deployment
Sprint 9	Maintenance & Continuous Improvement

4. Programming Language Selection

Backend

- Java (Spring Boot)

Reasons

- Secure
- Scalable
- Suitable for enterprise-level applications
- REST API support
- Excellent database connectivity
- Strong support for transaction management

Frontend

- HTML5

- CSS3
- JavaScript
- React.js

Database

- MySQL

Other Technologies

- REST API
- JWT Authentication
- Stripe/Razorpay Payment Gateway
- Git
- Docker
- AWS Cloud

Multi-Vendor Features

- Seller-specific dashboard and role-based access
- Seller product and inventory management
- Seller order fulfillment and sales tracking
- Admin approval and monitoring of seller accounts/products

5. System Architecture

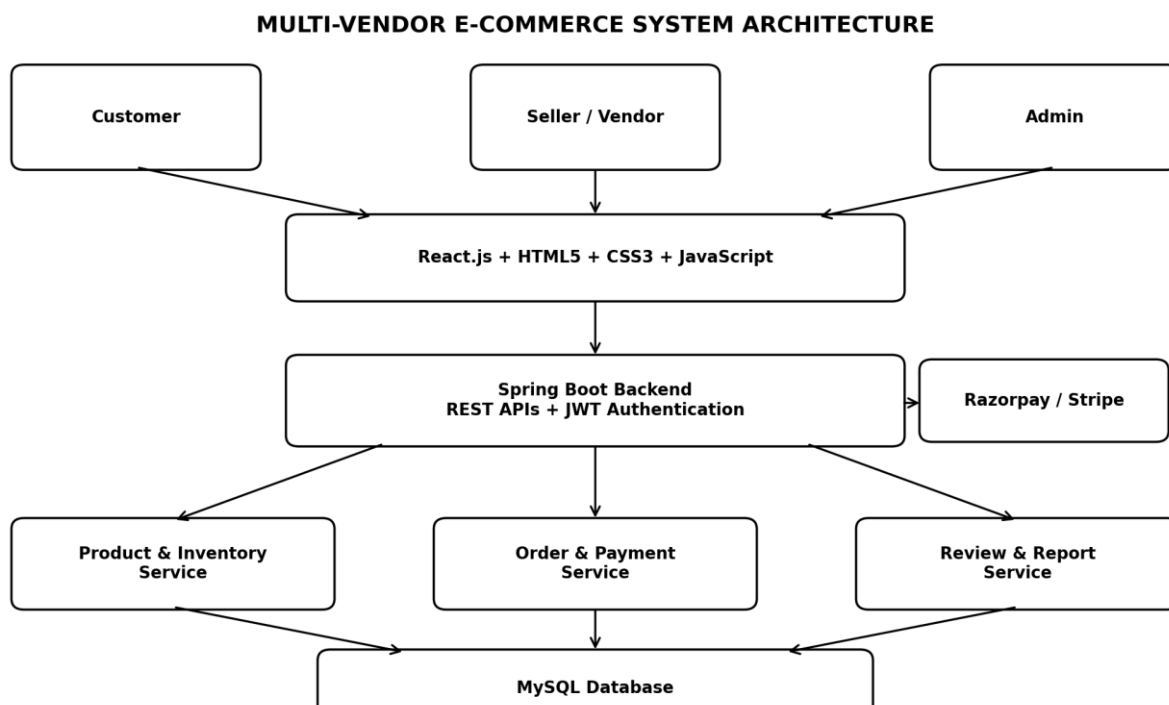


Figure 1: Multi-Vendor E-Commerce System Architecture

Architecture Explanation: Customers, sellers/vendors, and administrators access the React-based frontend. The Spring Boot backend provides REST APIs and JWT-based authentication. Separate services handle products and inventory, orders and payments, and reviews and reports. MySQL provides persistent storage, while Razorpay/Stripe is used for online payments.

Phase 1 – Requirement Analysis

- Define separate functional and access requirements for Customer, Seller, and Admin roles.
- Gather requirements from customers, sellers/vendors, and administrators.
- Gather requirements from customers, administrators, and business stakeholders.
- Create the Software Requirement Specification (SRS) document.
- Prepare use case diagrams for Customer and Admin modules.

Phase 2 – System Design

- Design database schema for sellers, products, inventory, orders, payments, and reviews.
- Design separate dashboards for customers, sellers, and administrators.
- Design database schema for users, products, categories, cart, orders, payments, and reviews.
- Create UI wireframes for product pages, cart, checkout, and admin dashboard.
- Develop UML diagrams (class, sequence, activity).
- Define system architecture using Spring Boot + React + MySQL.

Phase 3 – Development

- Implement seller registration, seller dashboard, and seller role-based access.
- Develop seller product listing, inventory, order fulfillment, and sales modules.
- Implement user authentication and role-based access control.
- Develop product catalog, search, cart, and wishlist modules.
- Build order management, checkout, and payment features.
- Integrate order tracking, reviews, and report generation.
- Create admin dashboard for product, inventory, order, and sales monitoring.

Phase 4 – Testing

- Unit Testing: Login, registration, product search, cart, and checkout.
- Integration Testing: Cart + Order placement; Payment + Order confirmation.
- System Testing: End-to-end e-commerce workflow validation.
- User Acceptance Testing (UAT): Verify customer and administrative requirements with real users.

Phase 5 – Deployment

- Verify seller workflows including product listing, inventory updates, order processing, and sales reporting.
- Deploy using Docker containers.
- Host the application on AWS EC2 for scalability.
- Use MySQL database for persistent storage.
- Configure Nginx web server for load balancing and security.

System Roles: Customer → purchases products; Seller/Vendor → lists and fulfills products; Admin → manages the marketplace and monitors all users.

7. Evaluation Strategy

Evaluation Parameter	Method
Functional Accuracy	Requirement verification and module validation
Performance	Load testing using JMeter
Security	Authentication, authorization, and penetration testing
Reliability	Continuous monitoring and backup validation
Usability	User feedback from customers and administrators
Scalability	Concurrent user testing under peak load
Maintainability	Code review, documentation, and version control (GitHub)

Seller Functionality	Validation of product listing, inventory, order fulfillment, and seller reporting
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10. Expected Outcomes

- Dedicated seller portal for managing products, stock, orders, and sales.
- Better coordination between customers, sellers, and administrators.
- Centralized product, order, inventory, and customer management.
- Secure and scalable e-commerce platform.
- Faster product search, checkout, and order processing.
- Reduced manual workload for administrators.
- Improved customer shopping experience.
- Reliable payment, data storage, and sales reporting.
- Continuous enhancement through Agile iterations.